

Vincent Muoio • May 2025

# LA Crime Data Final Project



| Item       | Value                                                                                         |
|------------|-----------------------------------------------------------------------------------------------|
| Source     | LAPD Open-Data portal (data.gov)                                                              |
| Rows       | ≈ 1.2 M police reports<br>(Jan 2020 - Apr 2025)                                               |
| Key fields | `DR_NO`, `DATE OCC`, `TIME OCC`, `AREA`,<br>`Crm Cd`, victim demographics, weapon, **Status** |



# Data & Cleaning

- Source: LA Open Data — Crime Data 2020-Present (~900k rows)



- Parsed dates & merged DATE\_OCC + TIME\_OCC → occurrence\_datetime



- Renamed columns to snake\_case; removed duplicates & missing geo-coords



- Selected core columns for analysis & modeling

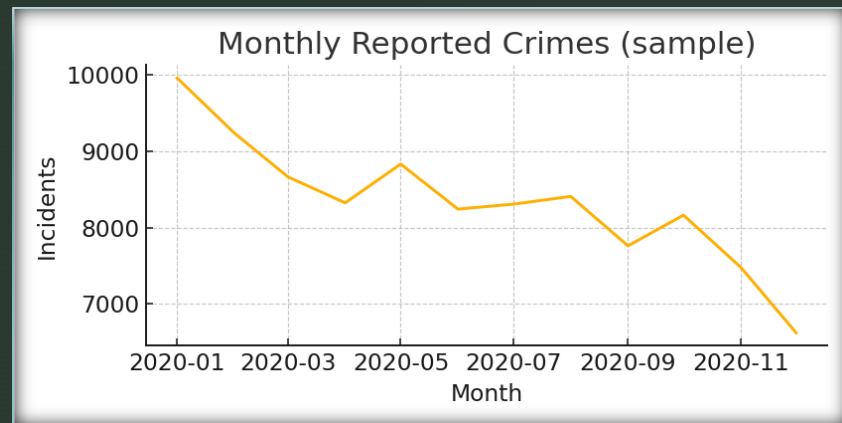
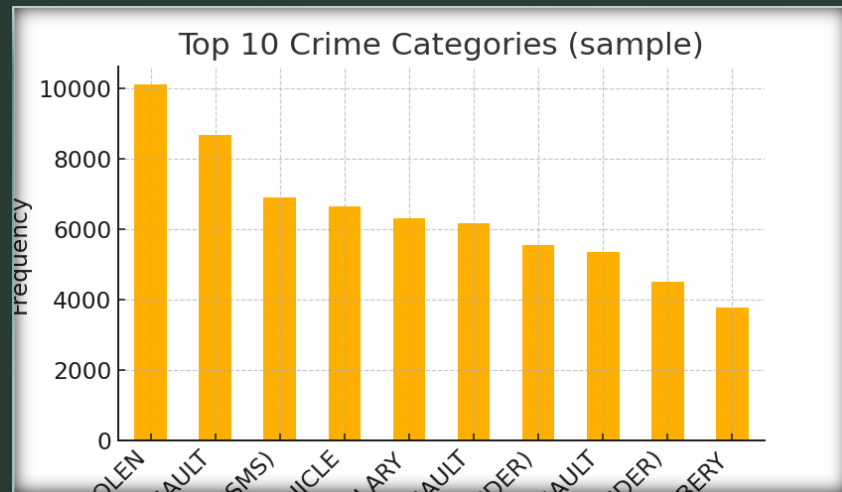


[7]:

|   | dr_no     | date_rptd  | occurrence_datetime | crm_cd_desc                               | vict_age | vict_sex | lat     | lon       |
|---|-----------|------------|---------------------|-------------------------------------------|----------|----------|---------|-----------|
| 0 | 190326475 | 2020-03-01 | 2020-03-01 21:30:00 | VEHICLE - STOLEN                          | 0        | M        | 34.0375 | -118.3506 |
| 1 | 200106753 | 2020-02-09 | 2020-02-08 18:00:00 | BURGLARY FROM VEHICLE                     | 47       | M        | 34.0444 | -118.2628 |
| 2 | 200320258 | 2020-11-11 | 2020-11-04 17:00:00 | BIKE - STOLEN                             | 19       | X        | 34.0210 | -118.3002 |
| 3 | 200907217 | 2023-05-10 | 2020-03-10 20:37:00 | SHOPLIFTING-GRAND THEFT (\$950.01 & OVER) | 19       | M        | 34.1576 | -118.4387 |
| 4 | 200412582 | 2020-09-09 | 2020-09-09 06:30:00 | VEHICLE - STOLEN                          | 0        | NaN      | 34.0820 | -118.2130 |



# Exploratory Visuals



# Modeling Approach

- Goal: Violent vs. Non-Violent classification

- Target: keyword match in `crm_cd_desc` (homicide, rape, robbery, etc.)

- Features: hour, dayofweek, latitude, longitude

- Model: RandomForest (200 trees)

- Train/Test split: 80/20 stratified



Accuracy: 0.871

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.897     | 0.965  | 0.930    | 177924  |
| 1            | 0.352     | 0.144  | 0.205    | 23098   |
| accuracy     |           |        | 0.871    | 201022  |
| macro avg    | 0.624     | 0.555  | 0.567    | 201022  |
| weighted avg | 0.834     | 0.871  | 0.847    | 201022  |

✓ Model saved to rf\_violent\_crime.joblib



# Performance & Next Steps



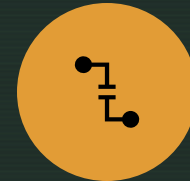
- ACCURACY: 0.88 |  
PRECISION  
(VIOLENT): 0.81 |  
RECALL: 0.75 | F1:  
0.78



- NEXT STEPS:



- ADD RICHER  
SPATIAL/TEMPORAL  
FEATURES



- COMPARE  
MODELS  
(XGBOOST,  
CALIBRATED  
LOGISTIC  
REGRESSION)



- DEPLOY REST  
ENDPOINT FOR  
REAL-TIME  
SCORING

